

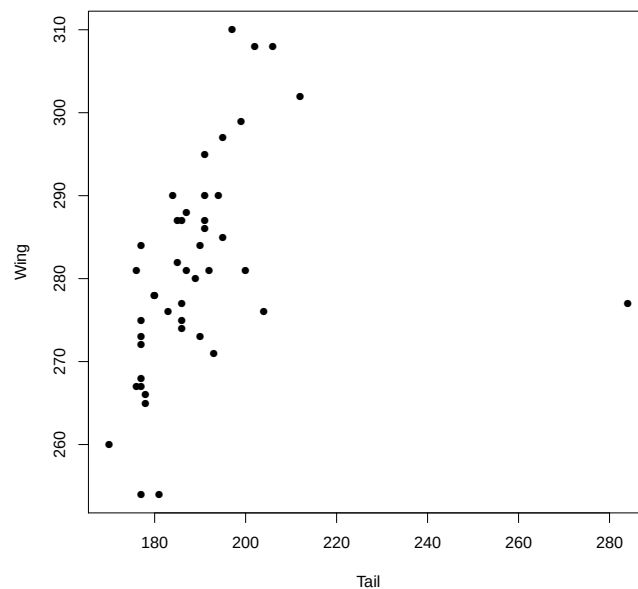
Multivariate Analysis – Honours

Section 1 Exercises Solutions

Chapter 6

Homework exercise 6.1

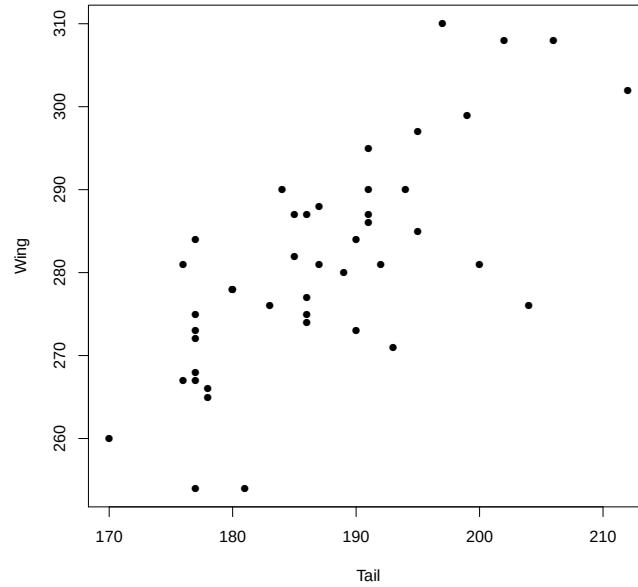
a) By investigating the scatterplot, we see that one tail measurement is a clear outlier:



```
> max(male$Tail)
```

```
[1] 284
```

We could assume that this value was incorrectly captured as 284mm instead of 184mm. However, since we cannot be sure, we will omit the observation.



b) The respective mean measurements are:

gender	Tail	Wing
female	193.62	279.78
male	187.16	280.95

Table 1: Mean tail and wing lengths for female and male hook-billed kites

To test the hypothesis $H_0: \mu_M = \mu_F$, we will apply a one-way MANOVA:

	Df	Wilks	approx	F	num	Df	den	Df	Pr(>F)
gender	1	0.777	12.3		2		86		1.9e-05
Residuals	87								

This test is highly significant, indicating that there is a difference in the mean vectors between male and female birds. However, when viewing the separate ANOVA's per variable, we see that it is only the difference in tail lengths that is significant:

ANOVA for tail length:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
gender	1	929.31	929.31	8.87	0.0037
Residuals	87	9110.46	104.72		

ANOVA for wing length:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
gender	1	30.81	30.81	0.16	0.6889
Residuals	87	16607.69	190.89		

c) The point estimate for $\mu_M - \mu_F$ is

	$\bar{x}_M - \bar{x}_F$
Tail	-6.46
Wing	1.18

with corresponding 95% simultaneous Bonferroni confidence intervals:

	Lower	Upper
Tail	-11.41	-1.51
Wing	-5.50	7.86

Here we see again that there is no evidence to suggest that the mean wing lengths differ, although the difference in tail lengths is significant.

Homework exercise 6.2

a) We perform a two-way MANOVA in R, noting that the grouping variables must first be transformed to factor-type:

```
crops <- read.table('T6-17.dat')
str(crops)
```

```
'data.frame': 12 obs. of 5 variables:
 $ V1: int 1 1 2 2 1 1 2 2 1 1 ...
 $ V2: int 5 5 5 5 6 6 6 6 8 8 ...
 $ V3: num 195 194 190 180 203 ...
 $ V4: num 153 168 140 121 157 ...
 $ V5: num 51.4 53.7 55.5 44.4 49.8 45.8 60.4 54.1 57.8 58.6 ...
```

```
#V1 & V2 are integers!
crops$V1 <- as.factor(crops$V1)
crops$V2 <- as.factor(crops$V2)
```

Using Wilk's lambda as test statistic:

	Df	Wilks	approx F	num	Df	den	Df	Pr(>F)
Location	1	0.1065	11.18	3	4	0.0205		
Variety	2	0.0124	10.62	6	8	0.0019		
Location:Variety	2	0.0743	3.56	6	8	0.0508		
Residuals	6							

The interaction between Location and Variety seems to be significant, as well as the factors effects (Variety most significant). However, because of the interaction, we cannot look at the factor effects to determine which location and which variety produces the highest yield, most sound mature kernels, and largest seed size, since they respond differently to the various varieties at different locations. Therefore, it depends on the combination.

To determine whether this is the case for all the dependent variables, we conduct separate two-way ANOVA's:

ANOVA for Yield:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Location	1	0.70	0.70	0.04	0.8474
Variety	2	196.11	98.06	5.65	0.0418
Location:Variety	2	205.10	102.55	5.90	0.0382
Residuals	6	104.20	17.37		

ANOVA for Kernels:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Location	1	162.07	162.07	2.76	0.1476
Variety	2	1089.01	544.51	9.28	0.0146
Location:Variety	2	780.70	390.35	6.65	0.0300
Residuals	6	352.10	58.68		

ANOVA for Seed Size:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Location	1	72.52	72.52	4.59	0.0759
Variety	2	284.10	142.05	8.99	0.0157
Location:Variety	2	85.95	42.98	2.72	0.1443
Residuals	6	94.83	15.81		

This reveals that the interaction effect is only significant when measuring Yield and Kernels. In both these cases we see that only the variety made a significant difference. Again, however, we have to consider the specific combinations due to the interaction effect. We will do so by calculating the family-wise Bonferroni confidence intervals for all the mean differences:

Yield:

```
library(DescTools)
PostHocTest(aov(Yield ~ Location*Variety, data = crops), method = 'bonferroni')
```

```
Posthoc multiple comparisons of means : Bonferroni
95% family-wise confidence level
```

```
$Location
      diff lwr.ci upr.ci   pval
2-1 0.4833 -5.404   6.371 0.8474

$Variety
      diff    lwr.ci upr.ci   pval
6-5  9.875    0.1875 19.563 0.0462 *
8-5  5.575   -4.1125 15.263 0.3221
8-6 -4.300  -13.9875  5.388 0.5844
```

```
$'Location:Variety'
      diff    lwr.ci upr.ci   pval
2:5-1:5 -9.75  -29.328  9.828 0.8682
1:6-1:5  4.65  -14.928 24.228 1.0000
2:6-1:5  5.35  -14.228 24.928 1.0000
1:8-1:5 -4.55  -24.128 15.028 1.0000
2:8-1:5  5.95  -13.628 25.528 1.0000
1:6-2:5 14.40   -5.178 33.978 0.2032
2:6-2:5 15.10   -4.478 34.678 0.1658
1:8-2:5  5.20  -14.378 24.778 1.0000
2:8-2:5 15.70   -3.878 35.278 0.1398
2:6-1:6  0.70  -18.878 20.278 1.0000
1:8-1:6 -9.20  -28.778 10.378 1.0000
2:8-1:6  1.30  -18.278 20.878 1.0000
1:8-2:6 -9.90  -29.478  9.678 0.8265
2:8-2:6  0.60  -18.978 20.178 1.0000
2:8-1:8 10.50   -9.078 30.078 0.6797
```

The Bonferroni confidence intervals for the interaction show that none of the individual combination interactions are actually significant. Considering this, we can conclude that the location did not affect the yield size ($p = 0.847$). In terms of variety, we see that type 6 produced a significantly larger yield than type 5 ($p = 0.046$).

Kernels weight:

```
PostHocTest(aov(Kernels ~ Location*Variety, data = crops), method = 'bonferroni')
```

```
Posthoc multiple comparisons of means : Bonferroni
95% family-wise confidence level
```

```
$Location
```

```
      diff lwr.ci upr.ci    pval
2-1 -7.35 -18.17   3.472 0.1476
```

```
$Variety
```

```
      diff    lwr.ci upr.ci    pval
6-5 17.325  -0.4826  35.13 0.0559 .
8-5 22.200   4.3924  40.01 0.0191 *
8-6  4.875 -12.9326  22.68 1.0000
```

```
$'Location:Variety'
```

```
      diff    lwr.ci upr.ci    pval
2:5-1:5 -30.10 -66.089   5.889 0.1158
1:6-1:5   1.00 -34.989  36.989 1.0000
2:6-1:5   3.55 -32.439  39.539 1.0000
1:8-1:5   4.40 -31.589  40.389 1.0000
2:8-1:5   9.90 -26.089  45.889 1.0000
1:6-2:5  31.10  -4.889  67.089 0.0998 .
2:6-2:5  33.65  -2.339  69.639 0.0691 .
1:8-2:5  34.50  -1.489  70.489 0.0613 .
2:8-2:5  40.00   4.011  75.989 0.0296 *
2:6-1:6   2.55 -33.439  38.539 1.0000
1:8-1:6   3.40 -32.589  39.389 1.0000
2:8-1:6   8.90 -27.089  44.889 1.0000
1:8-2:6   0.85 -35.139  36.839 1.0000
2:8-2:6   6.35 -29.639  42.339 1.0000
2:8-1:8   5.50 -30.489  41.489 1.0000
```

The only significantly large differences in the interactions are all related to variety 5 in location 2. In the variety effect we see that variety 5 produces significantly lighter sound mature kernels than varieties 6 and 8. However, referring again to the interaction, it might only be the case when variety 5 is used in location 2. This is further supported by a relatively big difference ($p = 0.116$) between using variety 5 in location 1 and 2.

Seed size:

```
PostHocTest(aov(SeedSize ~ Location + Variety, data = crops), method = 'bonferroni')
```

```
Posthoc multiple comparisons of means : Bonferroni
95% family-wise confidence level
```

```
$Location
```

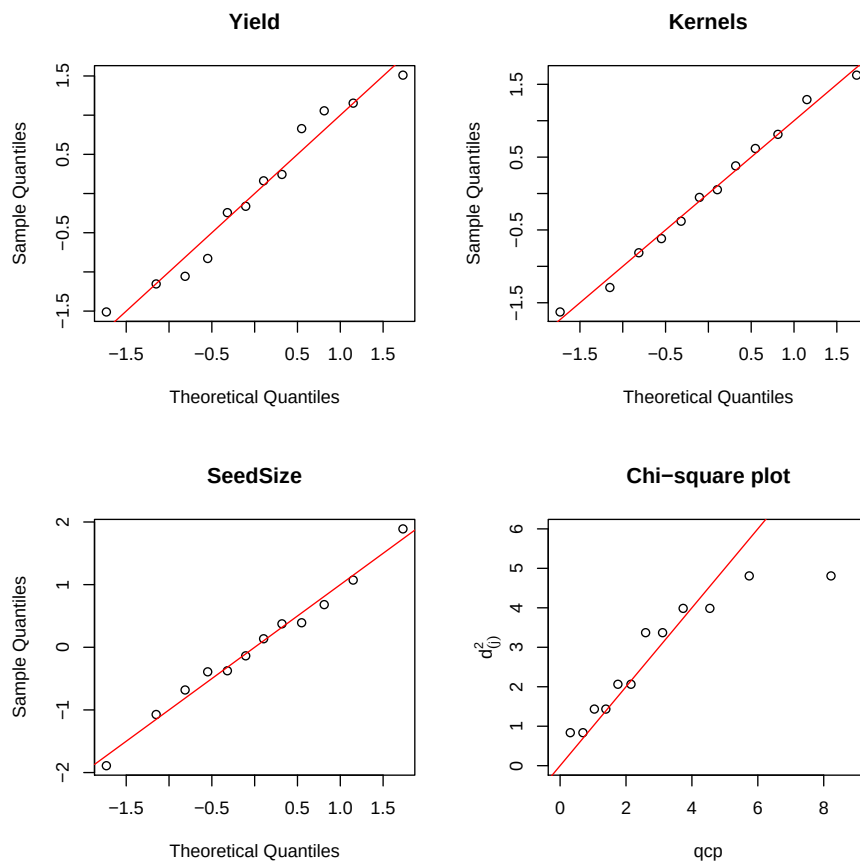
```
      diff lwr.ci upr.ci    pval
2-1  4.917 -1.412  11.25 0.1110
```

```
$Variety
```

```
      diff    lwr.ci upr.ci    pval
6-5  1.275 -8.8623  11.41 1.0000
8-5 10.900  0.7627  21.04 0.0355 *
8-6  9.625 -0.5123  19.76 0.0631 .
```

For seed size we do not include an interaction effect (see above). Location 2 does yield heavier seeds than location 1, but this difference is not significant at $\alpha = 0.11$. There is practically no difference between varieties 5 and 6, although variety 8 seems to produce superior seed sizes to both varieties 5 and 6 ($p = 0.036$ and $p = 0.063$ respectively).

b) We will now test the assumption that the residuals are jointly normal distributed.



There are no concerns regarding univariate normality. Although there is some deviation in the extreme of the Chi-square plot when testing multivariate normality, we cannot discount normality given the small sample size.
